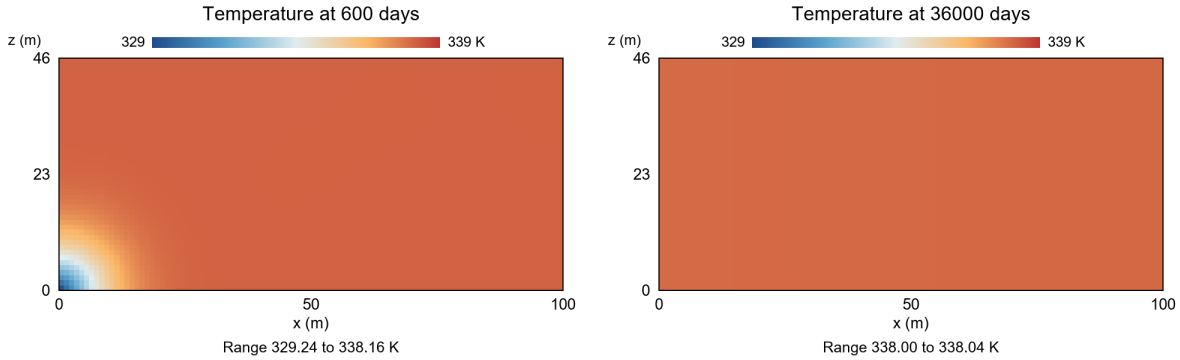
DARSim Example:

Thermal Compositional 2D CCS



This example demonstrates the Compositional_Energy model in DARSim for CO2 injection into a warmer brine reservoir. The model couples component mass balances with an energy balance, allowing pressure, composition and temperature to evolve together. The figures show the saved temperature fields at the end of injection, day 600 (left), and at day 36000, approximately 98.6 years (right). Both panels use the same color scale and an upward-increasing vertical coordinate.

The reservoir in Sim.txt is a 100 m by 1 m by 46 m Cartesian domain discretized into 100 by 1 by 46 cells. Porosity is 0.2. Perm_mSPE10T_v1.txt supplies the heterogeneous permeability in all three directions, in mD. Gravity, tabulated relative permeability and capillary pressure, and hysteresis are enabled. The relative-permeability and capillary-pressure curves are supplied through LookupTable.txt. Fractures, molecular diffusion and mechanical dispersion are disabled.

The initial state is brine at 25 MPa and 338.15 K (65 degrees Celsius), with no CO2. The fluid model has two phases and two components, using the CO2-Brine component set and a salinity of 100000 ppm. Rock density is 2600 kg/m3, rock specific heat is 850 J/(kg K), and rock thermal conductivity is 3 W/(m K). Energy transport includes fluid enthalpy transport, heat conduction and energy stored in the fluids and rock.

CO2 is injected at (0.5, 0.5, 0.5) m at 8e-5 pore volumes per day and 300 K. A pressure-controlled production well at (100, 0.5, 46) m is set to 25 MPa. Both wells operate from day 0 to day 600 and then close. In the saved results, the temperature range is 329.24 to 338.16 K at day 600 and 338.00 to 338.04 K at day 36000, showing local cooling during injection followed by thermal redistribution.

Select the thermal compositional model in Sim.txt. The two values below the model name specify the number of phases and components:

```
FLUID_MODEL
    Compositional_Energy
    2
    2
```

```
COMPONENT_LIST
    CO2-Brine
```

The initial reservoir temperature is set with INITIAL_TEMPERATURE. Inside the injection well definition, the temperature boundary condition and opening/closing schedule are:

```
BOUNDARY_CONDITION
    TEMPERATURE
    300
SCHEDULE
    0, 600
```